Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 3

Title: Building Dashboard for depicting Visualization and Deriving Insights

Submitted By:

Name: Prem Panchal

Reg. No.: 2411021240011

Date: 09/04/2026

GitHub Link: Prem2309/Business-Analytics-Projects

Submitted To:

Faculty Name: Mr. Aman Kumar Sharma

Department of Computer Application
Alliance University
Chandapura - Anekal Main Road, Anekal
Bengaluru – 562106

Experiment No.: 3

Building Dashboard for depicting Visualization and Deriving Insights

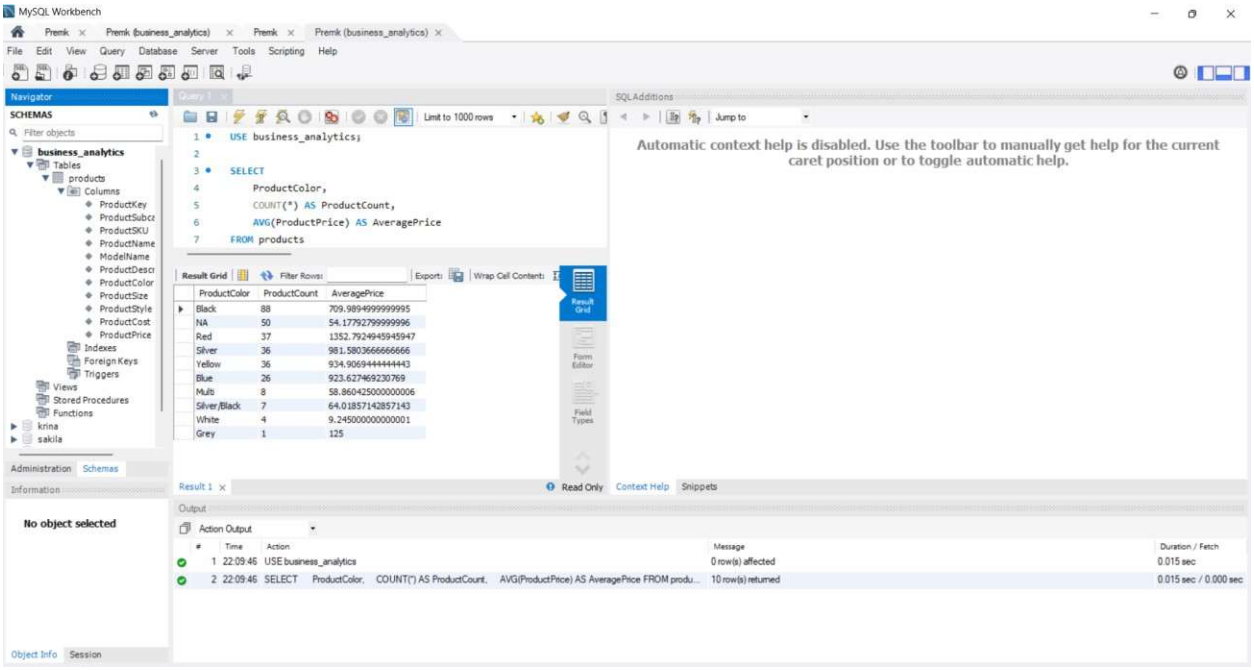
Problem Statement

You are a Data Analyst in a retail company. Sales data is stored in Excel files, customer data in CSV files, and product information in another source. Management wants a centralized reporting system. Import data from multiple sources into Power BI, establish the required connections, and prepare the data for business reporting.

Methodology / Steps Followed

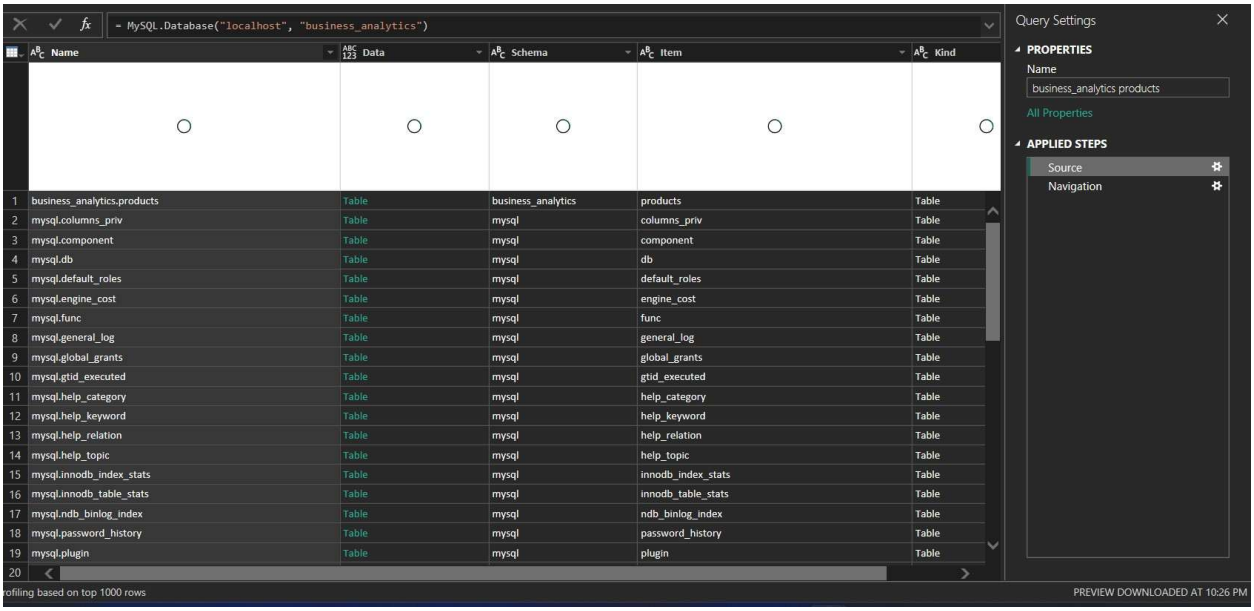
Step 1: Data Collection

- Obtained Sales Data from Excel files.
- Obtained Customer Data from CSV files.
- Obtained Product Data from another source (Excel/Database).
- First, I collected the required business datasets. Sales data was obtained from Excel files, customer data from CSV files, and product data from another source such as Excel or a database.
- The screenshot on page 2 also shows MySQL Workbench being used with a business analytics database, demonstrating the database side of the assignment.



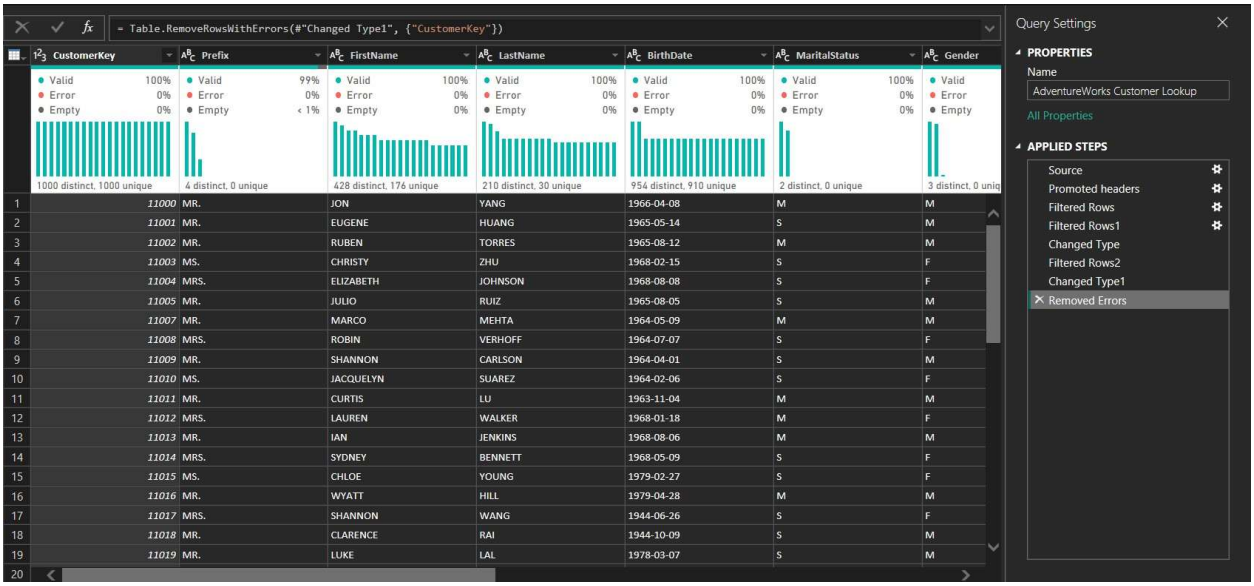
Step 2: Data Import

- Opened Power BI Desktop.
- Imported all datasets using the **Get Data** option.
- Loaded Excel and CSV files into Power BI.
- I opened Power BI Desktop and used the Get Data option to bring the datasets into Power BI. Excel and CSV data were loaded into the Power BI environment.
- The screenshot on page 3 shows the data source/navigation area with tables available from the database connection.



Step 3: Data Cleaning and Transformation

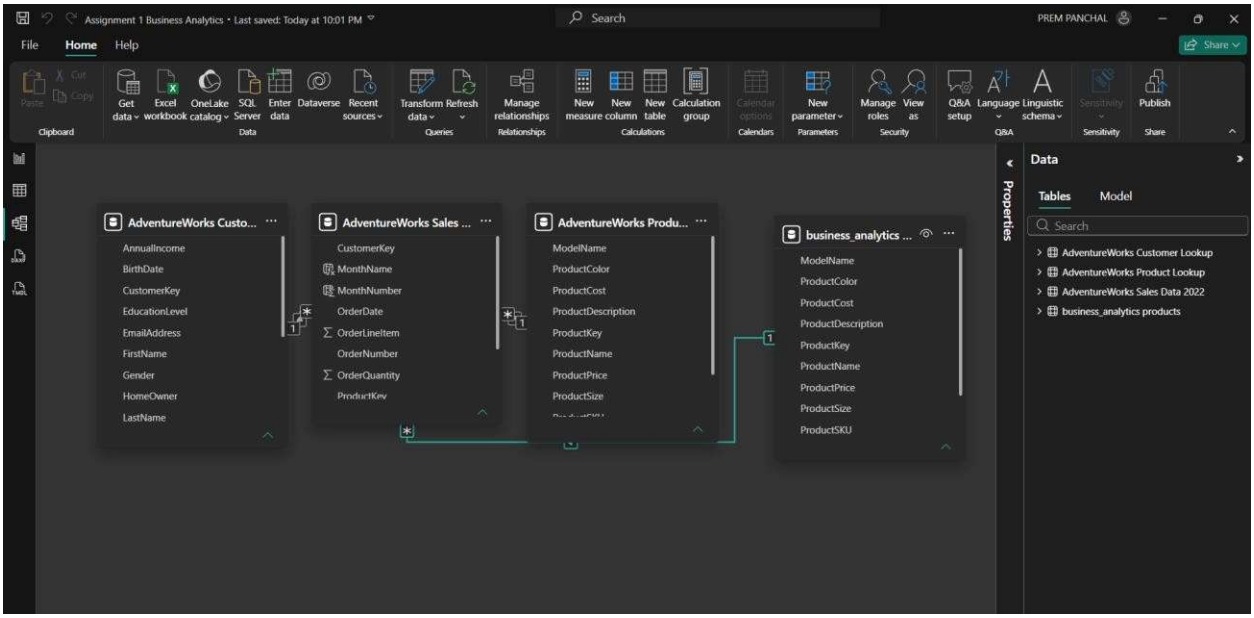
- Removed duplicate records.
- Handled missing values.
- Renamed columns appropriately.
- Converted data types where necessary.
- After importing the data,
- I prepared it for analysis.
- I removed duplicate records, handled missing values, renamed columns where required, and converted columns to appropriate data types.
- This work was performed during the data preparation stage using Power Query.



	CustomerKey	Prefix	FirstName	LastName	BirthDate	MaritalStatus	Gender
	Valid 100% Error 0% Empty 0%	Valid 99% Error 0% Empty < 1%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid Error Empty
	1000 distinct, 1000 unique	4 distinct, 0 unique	428 distinct, 176 unique	210 distinct, 30 unique	954 distinct, 910 unique	2 distinct, 0 unique	3 distinct, 0 uniq
1	11000	MR.	JON	YANG	1966-04-08	M	M
2	11001	MR.	EUGENE	HUANG	1965-05-14	S	M
3	11002	MR.	RUBEN	TORRES	1965-08-12	M	M
4	11003	MS.	CHRISTY	ZHU	1968-02-15	S	F
5	11004	MRS.	ELIZABETH	JOHNSON	1968-08-08	S	F
6	11005	MR.	JULIO	RUIZ	1965-08-05	S	M
7	11007	MR.	MARCO	MEHTA	1964-05-09	M	M
8	11008	MRS.	ROBIN	VERHOFF	1964-07-07	S	F
9	11009	MR.	SHANNON	CARLSON	1964-04-01	S	M
10	11010	MS.	JACQUELYN	SUAREZ	1964-02-06	S	F
11	11011	MR.	CURTIS	LU	1963-11-04	M	M
12	11012	MRS.	LAUREN	WALKER	1968-01-18	M	F
13	11013	MR.	IAN	JENKINS	1968-08-06	M	M
14	11014	MRS.	SYDNEY	BENNETT	1968-05-09	S	F
15	11015	MS.	CHLOE	YOUNG	1979-02-27	S	F
16	11016	MR.	WYATT	HILL	1979-04-28	M	M
17	11017	MRS.	SHANNON	WANG	1944-06-26	S	F
18	11018	MR.	CLARENCE	RAI	1944-10-09	S	M
19	11019	MR.	LUKE	LAL	1978-03-07	S	M
20							

Step 4: Data Modeling

- Created relationships between Sales, Customer, and Product tables.
- Verified one-to-many relationships.
- After cleaning the data, I created relationships between the Sales, Customer, and Product tables.
- I also verified the one-to-many relationships so that the tables could work together correctly during analysis.
- The model diagram shown on page 4 demonstrates these connections between the tables.

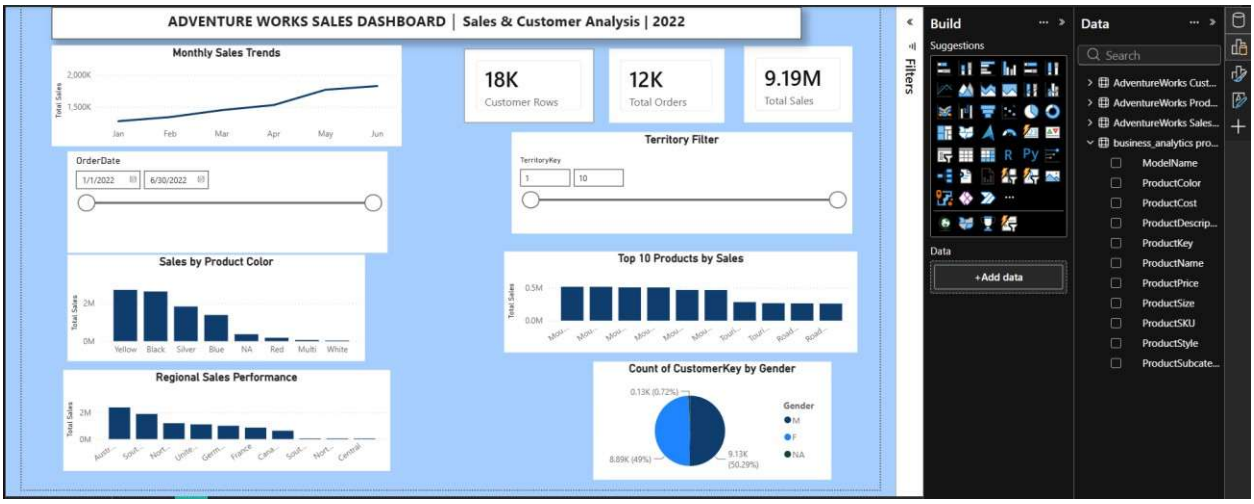


Step 5: Dashboard Development

- Created KPI cards for Total Sales, Orders, and Customers.
- Developed charts for Sales by Category and Region.
- Added slicers for Month, Category, and Region.
- Designed an interactive business dashboard.
- Finally, I created an interactive Power BI dashboard.
- I added KPI cards to display important measures such as total sales, total orders, and customers.
- I also created charts for sales analysis by category, region, product, and month. Slicers were added for filtering the report by month, category, and region.
- The dashboard screenshots on page 5 show these visualizations and interactive filters.

Output

The final Power BI dashboard provides:





- Total Sales Performance
- Customer Analysis
- Product-wise Sales Analysis
- Regional Sales Performance
- Monthly Sales Trends
- Interactive Filters and Slicers

The dashboard enables management to monitor business performance and make data-driven decisions effectively.

Conclusion

This project successfully integrated data from multiple sources into Power BI and transformed it into meaningful business insights. Data cleaning, transformation, and modeling techniques were applied to ensure accurate reporting. The developed dashboard provides a centralized view of business performance and supports better decision-making.

Learning Outcomes

Through this project, I learned:

- How to import data from Excel and CSV files into Power BI.
- Data cleaning and transformation using Power Query.
- Creating relationships between multiple tables.
- Developing DAX measures and calculated columns.
- Designing interactive dashboards and reports.
- Generating business insights from raw data.